

Bloch

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Contents

1	Construction of the Real Numbers	2
1.1	Axioms for the Natural numbers	2
1.1.1	Peano Postulates	2
1.1.2	Consequences of Peano Postulates	3
1.1.3	Addition on $\mathbb{N}$	6
1.1.4	Multiplication on $\mathbb{N}$	8
1.1.5	Properties of addition and multiplication on the naturals	11

Chapter 1

Construction of the Real Numbers

Definition 1.1. Let S be a set. A **binary operation** on S is a function $S \times S \rightarrow S$. A **unary operation** on S is a function $S \rightarrow S$.

Let S be a set and let $*$: $S \times S \rightarrow S$ be a binary operation. If $x, y \in S$, the correct way to write the result of the operation to the pair $\langle x, y \rangle$ would be $*(\langle x, y \rangle)$. However since this is a bit cumbersome one may write $x * y$ instead. Similarly if $\neg$: $S \rightarrow S$ is a unary operation and $x \in S$ one may choose to write $\neg x$ over $\neg(x)$.

1.1 Axioms for the Natural numbers

1.1.1 Peano Postulates

Axiom: Peano Postulates

There exists a set $\mathbb{N}$ with an element $1 \in \mathbb{N}$ and a function $s : \mathbb{N} \rightarrow \mathbb{N}$ that satisfy the following three properties.

- There is no $n \in \mathbb{N}$ such that $s(n) = 1$.
- The function s is injective.
- Let $G \subseteq \mathbb{N}$ be a set. Suppose that $1 \in G$, and that if $g \in G$ then $s(g) \in G$. Then $G = \mathbb{N}$.

It will be shown that the set $\mathbb{N}$ is unique. We can therefore give it a name.

Definition 1.2. The set of **natural numbers** denoted $\mathbb{N}$, is the set the existence of which is given in the Peano Postulates.

1.1.2 Consequences of Peano Postulates

Lemma 1.3. *Let $a \in \mathbb{N}$. Suppose that $a \neq 1$. Then there is a unique $b \in \mathbb{N}$ such that $a = s(b)$. (where existence of s and $\mathbb{N}$ are outlined by the peano axioms)*

Proof. First we prove existence. Let

$$G = \{1\} \cup \{c \in \mathbb{N} \mid \exists b \in \mathbb{N}(s(b) = c)\}$$

(which exists by the subset, pairing, and union axioms) We will prove that $G = \mathbb{N}$. To do this we need to show $1 \in G$, $G \subseteq \mathbb{N}$ and $\forall g \in G(s(g) \in G)$. The first two are straightforward. For the third suppose $g \in G$; we have two cases. First suppose $g \in \{1\}$; then $g = 1 \in \mathbb{N}$. Since $s : \mathbb{N} \rightarrow \mathbb{N}$, we then have $s(1) \in \mathbb{N}$. So by definition $s(1) \in \{c \in \mathbb{N} \mid \exists b \in \mathbb{N}(s(b) = c)\}$ and $s(g) = s(1) \in G$. Now suppose $g \in \{c \in \mathbb{N} \mid \exists b \in \mathbb{N}(s(b) = c)\}$; then by definition $g \in \mathbb{N}$ and $s(g) \in \mathbb{N}$ exists. By definition $s(g) \in \{c \in \mathbb{N} \mid \exists b \in \mathbb{N}(s(b) = c)\}$ so $s(g) \in G$. By the third peano postulate we then can assert that $G = \mathbb{N}$. So $a \in \mathbb{N} = G$. Since $a \neq 1$ we must then have $a \in \{c \in \mathbb{N} \mid \exists b \in \mathbb{N}(s(b) = c)\}$ and by definition, $\exists b \in \mathbb{N}(s(b) = a)$.

Now for uniqueness. Let m and n be arbitrary elements of $\mathbb{N}$ such that $a = s(m)$ and $a = s(n)$. Then $\langle m, a \rangle \in s$ and $\langle n, a \rangle \in s$. Since s is injective we must have $n = m$. $\square$

Theorem 1.4. *Let H be a set, Let $e \in H$ and let $k : H \rightarrow H$ be a function. Then there is a unique function $f : \mathbb{N} \rightarrow H$ such that $f(1) = e$ and $f \circ s = k \circ f$.*

Proof. Letting H, e , and k be arbitrary such that $e \in H$ and $k : H \rightarrow H$. We will show existence and uniqueness separately. We start with uniqueness.

Let g and h be arbitrary such that $g : \mathbb{N} \rightarrow H$, $g(1) = e$, and $g \circ s = k \circ g$ and $h : \mathbb{N} \rightarrow H$, $h(1) = e$, and $h \circ s = k \circ h$. We will first define V as the unique set (guaranteed by the subset and extensionality axioms) such that

$$\forall a(a \in V \leftrightarrow a \in \mathbb{N} \wedge g(a) = h(a))$$

and show that $V = \mathbb{N}$. By peano's postulates, we need to show $V \subseteq \mathbb{N}$, $1 \in V$, and $\forall n \in V(s(n) \in V)$. The first is straightforward. For the second requirement, we know by peano's postulates that $1 \in \mathbb{N}$, so by our assumptions $g(1)$ and $h(1)$ exist and $g(1) = e = h(1)$. So $1 \in V$.

For the third requirement, let n be arbitrary and suppose that $n \in V$. Then by definition $n \in \mathbb{N}$ and $g(n) = h(n)$. Since $n \in \mathbb{N} = \text{dom}(h \circ s)$, we have, by our assumptions,

$$h(s(n)) = h \circ s(n) = k \circ h(n) = k(h(n)) = k(g(n)) = k \circ g(n) = g \circ s(n) = g(s(n))$$

we know by definition $\text{ran}(s) \subseteq \mathbb{N}$, so $s(n) \in \mathbb{N}$ and again by definition $s(n) \in V$. We can conclude that $V = \mathbb{N}$.

Suppose $\langle x, y \rangle \in g$, then $x \in \text{dom } g = \mathbb{N} = V$. So by definition $y = g(x) = h(x)$, and $\langle x, y \rangle \in h$. Suppose $\langle x, y \rangle \in h$, then $x \in \text{dom } h = \mathbb{N} = V$. So by definition $y = h(x) = g(x)$, and $\langle x, y \rangle \in g$.

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Now for existence. We need to come up with a function from $\mathbb{N}$ into H such that $f(1) = e$ and $f \circ s = k \circ f$. We first define C , where

$$\forall X (X \in C \leftrightarrow X \subseteq \mathbb{N} \times H \wedge \langle 1, e \rangle \in X \wedge \forall n \forall y (\langle n, y \rangle \in X \rightarrow \langle s(n), k(y) \rangle \in X))$$

(guaranteed by the subset and extensionality axioms.) We propose that $f = \bigcap C$ is the desired function. To ensure its existence, we need to show that C is nonempty. We will show that $\mathbb{N} \times H \in C$. For the first requirement, clearly $\mathbb{N} \times H \subseteq \mathbb{N} \times H$. Next, since $1 \in \mathbb{N}$ and $e \in H$ then $\langle 1, e \rangle \in \mathbb{N} \times H$. Finally, let n and y be arbitrary such that $\langle n, y \rangle \in \mathbb{N} \times H$. Since $s : \mathbb{N} \rightarrow \mathbb{N}$ and $k : H \rightarrow H$, we must then have $s(n) \in \mathbb{N}$ and $k(y) \in H$. So $\langle s(n), k(y) \rangle \in \mathbb{N} \times H$. $f = \bigcap C$ is therefore a valid candidate.

Before we prove the functionality of f , we first state a few facts about f . First we note that $f \in C$. Let arbitrary $x \in \bigcap C$. Since C is nonempty there exists $X \in C$, and by definition $x \in X \subseteq \mathbb{N} \times H$. So $\bigcap C \subseteq \mathbb{N} \times H$. Next let arbitrary $X \in C$. By definition $\langle 1, e \rangle \in X$, so $\langle 1, e \rangle \in \bigcap C$. Finally let n and y be arbitrary and suppose $\langle n, y \rangle \in \bigcap C$. We need to show that $\langle s(n), k(y) \rangle \in \bigcap C$. Let arbitrary $X \in C$. We have $\langle n, y \rangle \in X$. Since $X \in C$, by definition we must then have $\langle s(n), k(y) \rangle \in X$. We are done.

Next we note that $\forall X \in C (f \subseteq X)$. This is straightforward to prove. Finally we show

$$\begin{aligned} \forall n \forall y ([\langle n, y \rangle \in f \wedge \langle n, y \rangle \neq \langle 1, e \rangle] \\ \rightarrow \exists m \exists u (\langle m, u \rangle \in f \wedge \langle n, y \rangle = \langle s(m), k(u) \rangle)) \end{aligned}$$

Let n and y be arbitrary such that $\langle n, y \rangle \in f$ and $\langle n, y \rangle \neq \langle 1, e \rangle$. Suppose, for contradiction, that the statement after the implication is false. Then $\forall m \forall u (\langle m, u \rangle \in f \rightarrow \langle n, y \rangle \neq \langle s(m), k(u) \rangle)$. The set $f - \{\langle n, y \rangle\}$ exists by the subset axiom; we will show that this set is a member of C . We know that $f \in C$, so $f \subseteq \mathbb{N} \times H$, and it is straightforward to show that $(f - \{\langle n, y \rangle\}) \subseteq \mathbb{N} \times H$. Since $\langle 1, e \rangle \neq \langle n, y \rangle$ and $\langle 1, e \rangle \in f$, we also have $\langle 1, e \rangle \in f - \{\langle n, y \rangle\}$. Finally, let arbitrary $\langle a, b \rangle \in f - \{\langle n, y \rangle\}$. Then $\langle a, b \rangle \in f \in C$ and $\langle s(a), k(b) \rangle \in f$. Since $\langle a, b \rangle \in f$, we must have, by our assumption, $\langle n, y \rangle \neq \langle s(a), k(b) \rangle$ and $\langle s(a), k(b) \rangle \in f - \{\langle n, y \rangle\}$. So $f - \{\langle n, y \rangle\} \in C$. It is straightforward to show that $f \not\subseteq f - \{\langle n, y \rangle\}$, contradicting the fact that $\forall X \in C (f \subseteq X)$.

We now prove functionality. We will do this by defining G :

$$\forall z (z \in G \leftrightarrow z \in \mathbb{N} \wedge \exists! x \in H (\langle z, x \rangle \in f))$$

(which exists by the subset axiom) and using the peano postulates to show that $G = \mathbb{N}$.

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We defined G :

$$\forall z(z \in G \leftrightarrow z \in \mathbb{N} \wedge \exists! x \in H(\langle z, x \rangle \in f))$$

To show that $G = \mathbb{N}$, we need to show that $G \subseteq \mathbb{N}$, $1 \in G$, and $\forall n \in G(s(n) \in G)$. The first requirement is straightforward. For the second, we already know by definition that $1 \in \mathbb{N}$. Since $f \in C$ we know by definition that $\langle 1, e \rangle \in f$, where $e \in H$. Let arbitrary $x' \in H$ such that $\langle 1, x' \rangle \in f$. Suppose for contradiction that $e \neq x'$. Then we have $\langle 1, e \rangle \neq \langle 1, x' \rangle$; so by the third statement proven earlier there exists m, u such that $\langle m, u \rangle \in f$ and $\langle 1, x' \rangle = \langle s(m), k(u) \rangle$. This means that $1 = s(m)$, where since $f \subseteq \mathbb{N} \times H$, $m \in \mathbb{N}$. This contradicts the peano postulates, which assert that $\neg \exists n \in \mathbb{N}(s(n) = 1)$.

For the third and final requirement, let n be an arbitrary element of G . Then by definition $n \in \mathbb{N}$ and there exists a unique x such that $\langle n, x \rangle \in f$. By definition we know $s(n) \in \mathbb{N}$, and since $f \in C$, we have $\langle s(n), k(x) \rangle \in f$, this verifies existence. Now let x' be arbitrary such that $\langle s(n), x' \rangle \in f$. By peano's postulates we know that $s(n) \neq 1$, so $\langle s(n), x' \rangle \neq \langle 1, e \rangle$. By the third statement proven earlier there exists m, u such that $\langle m, u \rangle \in f$ and $\langle s(n), x' \rangle = \langle s(m), k(u) \rangle$. So $s(n) = s(m)$ and $x' = k(u)$. Since s is injective we must have $m = n$. So $\langle n, u \rangle \in f$. By uniqueness of x we then have $u = x$. So $x' = k(u) = k(x)$, proving uniqueness. $s(n) \in G$.

We have shown that $G = \mathbb{N}$. It follows from this that $f : \mathbb{N} \rightarrow H$. For functionality, let $x \in \text{dom}f$. Then there exists y such that $\langle x, y \rangle \in f$. Since $f \subseteq \mathbb{N} \times H$ we have $x \in \mathbb{N} = G$, and by definition y is unique. Next to show $\text{dom}f = \mathbb{N}$, first let arbitrary $x \in \text{dom}f$, similarly it immediately follows that $x \in \mathbb{N}$. Now let $x \in \mathbb{N}$. Then $x \in G$ and by definition there exists $y \in H$ such that $\langle x, y \rangle \in f$, meaning $x \in \text{dom}f$. Finally to show $\text{ran}f \subseteq H$, let $y \in \text{ran}f$. Then there exists x such that $\langle x, y \rangle \in f \subseteq \mathbb{N} \times H$, so $y \in H$.

The fact that $\langle 1, e \rangle \in \bigcap C = f$ was already shown earlier. Finally we show $f \circ s = k \circ f$. Let $\langle x, y \rangle \in f \circ s$. Then $y = f \circ s(x) = f(s(x))$. Since $s(x) \neq 1$, $\langle s(x), y \rangle \neq \langle 1, e \rangle$ and as proven earlier there then exists m, u such that $\langle m, u \rangle \in f$ and $\langle s(x), y \rangle = \langle s(m), k(u) \rangle$. So $s(x) = s(m)$ and $y = k(u)$. Since s is injective $x = m$. We then have $\langle x, u \rangle \in f$ and $y = k(u) = k(f(x)) = k \circ f(x)$. so $\langle x, y \rangle \in k \circ f$. Next, suppose $\langle x, y \rangle \in k \circ f$. By definition there exists z such that $\langle x, z \rangle \in f$ and $\langle z, y \rangle \in k$. Since $f \in C$ we have $\langle s(x), k(z) \rangle \in f$. We then have $y = k(z) = f(s(x)) = f \circ s(x)$ and $\langle x, y \rangle \in f \circ s$. We are done. $\square$

1.1.3 Addition on $\mathbb{N}$

Theorem 1.5. *There is a unique binary operation $+$: $\mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$ that satisfies the following two properties for all $n, m \in \mathbb{N}$:*

- $n + 1 = s(n)$
- $n + s(m) = s(n + m)$

Proof. We start with uniqueness. Let $+$ and $\oplus$ be arbitrary such that $+, \oplus : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$, $\forall m \in \mathbb{N} \forall n \in \mathbb{N} (+(\langle n, 1 \rangle) = s(n) \wedge +(\langle n, s(m) \rangle) = s(+(\langle n, m \rangle)))$, and $\forall m \in \mathbb{N} \forall n \in \mathbb{N} (\oplus(\langle n, 1 \rangle) = s(n) \wedge \oplus(\langle n, s(m) \rangle) = s(\oplus(\langle n, m \rangle)))$. We need to show $+$ = $\oplus$. We define G as

$$\forall x (x \in G \leftrightarrow x \in \mathbb{N} \wedge \forall n \in \mathbb{N} (+(\langle n, x \rangle) = \oplus(\langle n, x \rangle)))$$

(guaranteed by the subset axiom.) We will show that $G = \mathbb{N}$. To do this we need to show $G \subseteq \mathbb{N}$, $1 \in G$, and $\forall g \in G (s(g) \in G)$. The first requirement is straightforward. For the second requirement, we already know $1 \in \mathbb{N}$, and for arbitrary $n \in \mathbb{N}$, $+(\langle n, 1 \rangle) = s(n) = \oplus(\langle n, 1 \rangle)$.

For the third requirement, let arbitrary $g \in G$. Then by definition $g \in \mathbb{N}$ and $\forall n \in \mathbb{N} (+(\langle n, g \rangle) = \oplus(\langle n, g \rangle))$. Since $s : \mathbb{N} \rightarrow \mathbb{N}$ we know that $s(g) \in \mathbb{N}$ exists. Let n be arbitrary. By definition $+(\langle n, s(g) \rangle) = s(+(\langle n, g \rangle))$ and $\oplus(\langle n, s(g) \rangle) = s(\oplus(\langle n, g \rangle))$. By assumption we can assert that $+(\langle n, g \rangle) = \oplus(\langle n, g \rangle)$, so $+(\langle n, s(g) \rangle) = s(+(\langle n, g \rangle)) = s(\oplus(\langle n, g \rangle)) = \oplus(\langle n, s(g) \rangle)$. We conclude that $s(g) \in G$ and $G = \mathbb{N}$.

Let $\langle \langle x, y \rangle, z \rangle \in +$. By definition $y \in \mathbb{N} = G$ and $x \in \mathbb{N}$. So $+(\langle x, y \rangle) = \oplus(\langle x, y \rangle)$ and $z = \oplus(\langle x, y \rangle)$. So $\langle \langle x, y \rangle, z \rangle \in \oplus$. Now let $\langle \langle x, y \rangle, z \rangle \in \oplus$. A similar argument shows that $\langle \langle x, y \rangle, z \rangle \in +$.

Now for existence. We want to show

$$\begin{aligned} \exists + (+ : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N} \wedge \forall m \in \mathbb{N} \forall n \in \mathbb{N} (&+(\langle n, 1 \rangle) = s(n) \\ &\wedge +(\langle n, s(m) \rangle) = s(+(\langle n, m \rangle)))) \end{aligned}$$

Theorem 1.4 allows us to assert

$$\forall k \forall H \forall e ([e \in H \wedge k : H \rightarrow H] \rightarrow \exists ! f (f : \mathbb{N} \rightarrow H \wedge f(1) = e \wedge f \circ s = k \circ f))$$

It immediately follows that

$$\forall p \in \mathbb{N} \exists ! f (f : \mathbb{N} \rightarrow \mathbb{N} \wedge f(1) = p \wedge f \circ s = s \circ f)$$

We will define $+$ as the set where

$$\begin{aligned} \forall z (z \in + \leftrightarrow \exists a \in \mathbb{N} \exists b \in \mathbb{N} (\exists f (f : \mathbb{N} \rightarrow \mathbb{N} \wedge f(1) = s(a) \wedge f \circ s = s \circ f \\ \wedge \exists c (\langle b, c \rangle \in f \wedge z = \langle \langle a, b \rangle, c \rangle)))) \end{aligned}$$

Existence follows from the subset axiom, where it can be shown that $+$ $\subseteq (\mathbb{N} \times \mathbb{N}) \times \mathbb{N}$.

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We defined $+$ as

$$\forall z(z \in + \leftrightarrow \exists a \in \mathbb{N} \exists b \in \mathbb{N} (\exists f(f : \mathbb{N} \rightarrow \mathbb{N} \wedge f(1) = s(a) \wedge f \circ s = s \circ f \wedge \exists c(\langle b, c \rangle \in f \wedge z = \langle \langle a, b \rangle, c \rangle))))$$

First we prove functionality. Let x be an arbitrary element of $\text{dom}(+)$. Then there exists y such that $\langle x, y \rangle \in +$. Let y' be arbitrary such that $\langle x, y' \rangle \in +$. $\langle x, y \rangle \in +$ by definition means that there exists $a, b \in \mathbb{N}$, and $f : \mathbb{N} \rightarrow \mathbb{N}$ such that $f(1) = s(a)$ and $f \circ s = s \circ f$, and that there exists c such that $\langle b, c \rangle \in f$ and $\langle x, y \rangle = \langle \langle a, b \rangle, c \rangle$. $\langle x, y' \rangle \in +$ similarly implies the existence of $a_0, b_0 \in \mathbb{N}$ and $f_0 : \mathbb{N} \rightarrow \mathbb{N}$ such that $f_0(1) = s(a_0)$ and $f_0 \circ s = s \circ f_0$, and the existence of c_0 such that $\langle b_0, c_0 \rangle \in f_0$ and $\langle x, y' \rangle = \langle \langle a_0, b_0 \rangle, c_0 \rangle$. Then $\langle a, b \rangle = x = \langle a_0, b_0 \rangle$, so $a_0 = a$ and $b_0 = b$. It follows that $s(a_0) = s(a) \in \mathbb{N}$. This and

$$\forall p \in \mathbb{N} \exists! f(f : \mathbb{N} \rightarrow \mathbb{N} \wedge f(1) = p \wedge f \circ s = s \circ f)$$

implies that $f = f_0$. So we have $y = c = f(b) = f(b_0) = f_0(b_0) = c_0 = y'$.

Next we show that $\text{dom}(+) = \mathbb{N} \times \mathbb{N}$. First let x be an arbitrary element of $\text{dom}(+)$. Then there exists y such that $\langle x, y \rangle \in +$. By definition there then exists $a, b \in \mathbb{N}$ and there exists c such that $\langle x, y \rangle = \langle \langle a, b \rangle, c \rangle$. So $x = \langle a, b \rangle \in \mathbb{N} \times \mathbb{N}$.

Now suppose $x \in \mathbb{N} \times \mathbb{N}$. Then by definition there exists $a, b \in \mathbb{N}$ such that $x = \langle a, b \rangle$. $s(a) \in \mathbb{N}$ exists, so by theorem 1.4 there exists f such that $f : \mathbb{N} \rightarrow \mathbb{N}$, $f(1) = s(a)$, and $f \circ s = s \circ f$. Since $b \in \mathbb{N} = \text{dom} f$, $f(b)$ exists and $\langle b, f(b) \rangle \in f$. Then by definition, $\langle \langle a, b \rangle, f(b) \rangle \in +$, so $x = \langle a, b \rangle \in \text{dom}(+)$.

Next we show that $\text{ran}(+) \subseteq \mathbb{N}$. Let arbitrary $y \in \text{ran}(+)$. Then there exists x such that $\langle x, y \rangle \in +$. By definition there then exists $a, b \in \mathbb{N}$, $f : \mathbb{N} \rightarrow \mathbb{N}$, and c such that $\langle b, c \rangle \in f$ and $\langle x, y \rangle = \langle \langle a, b \rangle, c \rangle$. So we have $y = c = f(b) \in \mathbb{N}$.

Now we verify the two properties of addition, that is,

$$\forall m \in \mathbb{N} \forall n \in \mathbb{N} (+(\langle n, 1 \rangle) = s(n) \wedge +(\langle n, s(m) \rangle) = s(+(\langle n, m \rangle)))$$

Let arbitrary $n, m \in \mathbb{N}$. We know that $+: \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$, so $\langle n, 1 \rangle \in \mathbb{N} \times \mathbb{N} = \text{dom}(+)$ and there exists y such that $\langle \langle n, 1 \rangle, y \rangle \in +$. By definition there then exists $a, b \in \mathbb{N}$ and $f : \mathbb{N} \rightarrow \mathbb{N}$ such that $f(1) = s(a)$, and there exists c such that $\langle b, c \rangle \in f$ and $\langle \langle n, 1 \rangle, y \rangle = \langle \langle a, b \rangle, c \rangle$. Then $n = a$, $1 = b$, and $y = c$. We then have $\langle 1, y \rangle \in f$ and $y = f(1) = s(a) = s(n)$. We conclude that $\langle \langle n, 1 \rangle, s(n) \rangle \in +$ and $+(\langle n, 1 \rangle) = s(n)$.

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Now for the second property. We know that $s(m) \in \mathbb{N}$. So $\langle n, s(m) \rangle \in \mathbb{N} \times \mathbb{N} = \text{dom}(+)$ and there exists y such that $\langle \langle n, s(m) \rangle, y \rangle \in +$. By definition there then exists $a, b \in \mathbb{N}$ such that $f : \mathbb{N} \rightarrow \mathbb{N}$, $f(1) = s(a)$, and $f \circ s = s \circ f$, and there exists c such that $\langle b, c \rangle \in f$ and $\langle \langle n, s(m) \rangle, y \rangle = \langle \langle a, b \rangle, c \rangle$. We then have $n = a$, $s(m) = b$, and $y = c = f(b) = f(s(m)) = s(f(m))$. We know that $\langle n, m \rangle \in \mathbb{N} \times \mathbb{N} = \text{dom}(+)$, so there exists y_0 such that $\langle \langle n, m \rangle, y_0 \rangle \in +$. Again by definition there exists $a_0, b_0 \in \mathbb{N}$ and $f_0 : \mathbb{N} \rightarrow \mathbb{N}$ such that $f_0(1) = s(a_0)$ and $f_0 \circ s = s \circ f_0$ and there exists c_0 such that $\langle b_0, c_0 \rangle \in f_0$ and $\langle \langle n, m \rangle, y_0 \rangle = \langle \langle a_0, b_0 \rangle, c_0 \rangle$. Since it follows that $a_0 = n = a$ and $s(a_0) = s(a)$, theorem 1.4 allows us to assert $f = f_0$. Putting it all together, we have $+(\langle n, m \rangle) = y_0 = c_0 = f_0(b_0) = f_0(m) = f(m)$. We conclude that $+(\langle n, s(m) \rangle) = y = s(f(m)) = s(+(\langle n, m \rangle))$. $\square$

1.1.4 Multiplication on $\mathbb{N}$

Theorem 1.6. *There is a unique binary operation $\cdot : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$ that satisfies the following two properties for all $n, m \in \mathbb{N}$:*

- $n \cdot 1 = n$
- $n \cdot s(m) = (n \cdot m) + n$

Proof. We start with uniqueness. Let $\cdot, \odot$ be arbitrary. Suppose $\cdot : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$ and $\forall m \in \mathbb{N} \forall n \in \mathbb{N} (\cdot(\langle n, 1 \rangle) = n \wedge \cdot(\langle n, s(m) \rangle) = +(\cdot(\langle n, m \rangle), n))$. Similarly suppose $\odot : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$ and $\forall m \in \mathbb{N} \forall n \in \mathbb{N} (\odot(\langle n, 1 \rangle) = n \wedge \odot(\langle n, s(m) \rangle) = +(\odot(\langle n, m \rangle), n))$. We want to show that $\cdot = \odot$. We do this by defining G such that

$$\forall z (z \in G \leftrightarrow z \in \mathbb{N} \wedge \forall n \in \mathbb{N} (\cdot(\langle n, z \rangle) = \odot(\langle n, z \rangle)))$$

(where existence is assured by the subset axiom) and showing that $G = \mathbb{N}$.

We need to show $G \subseteq \mathbb{N}$, $1 \in G$, and $\forall g \in G (s(g) \in G)$. The first requirement is straightforward. For the second, we already know $1 \in \mathbb{N}$, and see that for arbitrary $n \in \mathbb{N}$ we have $\cdot(\langle n, 1 \rangle) = n = \odot(\langle n, 1 \rangle)$.

For the third requirement let arbitrary $g \in G$. Then by definition $g \in \mathbb{N}$ and $\forall n \in \mathbb{N} (\cdot(\langle n, g \rangle) = \odot(\langle n, g \rangle))$. We know that $s(g) \in \mathbb{N}$ exists. So since $\langle n, s(g) \rangle \in \mathbb{N} \times \mathbb{N} = \text{dom}(\cdot)$, we have, by our assumptions, $\cdot(\langle n, s(g) \rangle) = +(\cdot(\langle n, g \rangle), n) = +(\odot(\langle n, g \rangle), n) = \odot(\langle n, s(g) \rangle)$. So $s(g) \in G$ and $G = \mathbb{N}$.

To prove the desired result, let arbitrary $z \in \cdot \subseteq (\mathbb{N} \times \mathbb{N}) \times \mathbb{N}$. Then there exists $a_0, a_1, b \in \mathbb{N}$ such that $\langle \langle a_0, a_1 \rangle, b \rangle = z \in \cdot$. So $b = \cdot(\langle a_0, a_1 \rangle)$. Since $a_1 \in \mathbb{N} = G$, we can then conclude that $b = \cdot(\langle a_0, a_1 \rangle) = \odot(\langle a_0, a_1 \rangle)$. So $z = \langle \langle a_0, a_1 \rangle, b \rangle \in \odot$. Next suppose $z \in \odot$. It then follows from a similar argument that $z \in \cdot$. We are done.

(next page)

Now for existence. We start by defining a function h for all q , where

$$\forall q \exists h \forall z (z \in h \leftrightarrow \exists m \in \mathbb{N} \exists x \in \mathbb{N} (z = \langle m, x \rangle \wedge \langle \langle m, q \rangle, x \rangle \in +))$$

(which exists by the subset axiom, where $\mathbb{N} \times \mathbb{N}$ is a bounding set. Uniqueness for each q follows from extensionality.) We show that for all $q \in \mathbb{N}$, this h is a function from $\mathbb{N}$ into $\mathbb{N}$. That is

$$\forall q \in \mathbb{N} [\forall z (z \in h \leftrightarrow \exists m \in \mathbb{N} \exists x \in \mathbb{N} (z = \langle m, x \rangle \wedge \langle \langle m, q \rangle, x \rangle \in +)) \rightarrow h : \mathbb{N} \rightarrow \mathbb{N}]$$

Let arbitrary $q \in \mathbb{N}$. Suppose that the statement before the implication is true. First to prove functionality. Let $a \in \text{dom}h$. Then there exists b such that $\langle a, b \rangle \in h$. Let b' be arbitrary such that $\langle a, b' \rangle \in h$. By our assumptions we must have $a, b, b' \in \mathbb{N}$, $+(\langle a, q \rangle) = b$, and $+(\langle a, q \rangle) = b'$. Since $+: \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$ we must have $b = b'$. Next for the domain bound suppose arbitrary $a \in \text{dom}h$. Then there exists b such that $\langle a, b \rangle \in h$. It immediately follows that $a \in \mathbb{N}$. Suppose instead that $a \in \mathbb{N}$. Then $\langle a, q \rangle \in \mathbb{N} \times \mathbb{N} = \text{dom}(+)$. So $+(\langle a, q \rangle) \in \mathbb{N}$ exists and by definition, $\langle a, +(\langle a, q \rangle) \rangle \in h$. So $a \in \text{dom}h$. Finally for the range bound, let arbitrary $b \in \text{ran}h$. Then there exists a such that $\langle a, b \rangle \in h$. It immediately follows by definition that $b \in \mathbb{N}$. We have therefore shown

$$\begin{aligned} \forall q \in \mathbb{N} \exists! h (\forall z (z \in h \leftrightarrow \exists m \in \mathbb{N} \exists x \in \mathbb{N} (z = \langle m, x \rangle \wedge \langle \langle m, q \rangle, x \rangle \in +)) \\ \wedge h : \mathbb{N} \rightarrow \mathbb{N}) \end{aligned}$$

Theorem 1.4 states

$$\forall H \forall q \forall h ([q \in H \wedge h : H \rightarrow H] \rightarrow \exists! g (g : \mathbb{N} \rightarrow H \wedge g(1) = q \wedge g \circ s = h \circ g))$$

which allows us to assert

$$\forall q \forall h ([q \in \mathbb{N} \wedge h : \mathbb{N} \rightarrow \mathbb{N}] \rightarrow \exists! g (g : \mathbb{N} \rightarrow \mathbb{N} \wedge g(1) = q \wedge g \circ s = h \circ g))$$

(next page)

We showed

$$\forall q \in \mathbb{N} \exists! h (\forall z (z \in h \leftrightarrow \exists m \in \mathbb{N} \exists x \in \mathbb{N} (z = \langle m, x \rangle \wedge \langle \langle m, q \rangle, x \rangle \in +)) \wedge h : \mathbb{N} \rightarrow \mathbb{N})$$

and

$$\forall q \forall h ([q \in \mathbb{N} \wedge h : \mathbb{N} \rightarrow \mathbb{N}] \rightarrow \exists! g (g : \mathbb{N} \rightarrow \mathbb{N} \wedge g(1) = q \wedge g \circ s = h \circ g))$$

Now we construct $\cdot$ as the set

$$\begin{aligned} \forall z (z \in \cdot \leftrightarrow \exists a \in \mathbb{N} \exists b \in \mathbb{N} (\\ \exists h (\forall z_0 (z_0 \in h \leftrightarrow \exists m \in \mathbb{N} \exists x \in \mathbb{N} (z_0 = \langle m, x \rangle \wedge \langle \langle m, a \rangle, x \rangle \in +)) \wedge h : \mathbb{N} \rightarrow \mathbb{N} \wedge \exists g (g : \mathbb{N} \rightarrow \mathbb{N} \wedge g(1) = a \wedge g \circ s = h \circ g \\ \wedge \exists c (\langle b, c \rangle \in g \wedge z = \langle \langle a, b \rangle, c \rangle)))))) \end{aligned}$$

(existence follows from the subset axiom, where $(\mathbb{N} \times \mathbb{N}) \times \mathbb{N}$ is a bounding set.)
We want to verify

$$\cdot : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N} \wedge \forall m \in \mathbb{N} \forall n \in \mathbb{N} (\cdot(\langle n, 1 \rangle) = n \wedge \cdot(\langle n, s(m) \rangle) = +(\cdot(\langle n, m \rangle), n))$$

We start with functionality. Let arbitrary $p \in \text{dom}(\cdot)$. Then by definition there exists q such that $\langle p, q \rangle \in \cdot$. This means that there exists $a, b \in \mathbb{N}$ and $h : \mathbb{N} \rightarrow \mathbb{N}$ such that $\forall z_0 (z_0 \in h \leftrightarrow \exists m \in \mathbb{N} \exists x \in \mathbb{N} (z_0 = \langle m, x \rangle \wedge \langle \langle m, a \rangle, x \rangle \in +))$. There also exists $g : \mathbb{N} \rightarrow \mathbb{N}$ such that $g(1) = a$ and $g \circ s = h \circ g$, and c such that $\langle b, c \rangle \in g$ and $\langle p, q \rangle = \langle \langle a, b \rangle, c \rangle$. Now suppose there exists $\langle p, q' \rangle \in \cdot$. Similarly there exists $a', b' \in \mathbb{N}$, $h' : \mathbb{N} \rightarrow \mathbb{N}$ such that $\forall z_0 (z_0 \in h' \leftrightarrow \exists m \in \mathbb{N} \exists x \in \mathbb{N} (z_0 = \langle m, x \rangle \wedge \langle \langle m, a' \rangle, x \rangle \in +))$, $g' : \mathbb{N} \rightarrow \mathbb{N}$ such that $g'(1) = a'$ and $g' \circ s = h' \circ g'$, and c' such that $\langle b', c' \rangle \in g'$ and $\langle p, q' \rangle = \langle \langle a', b' \rangle, c' \rangle$. We have $\langle a', b' \rangle = p = \langle a, b \rangle$. So $a' = a$ and $b' = b$. Uniqueness, as proven, implies that $h = h'$ and $g = g'$. So $q' = c' = g'(b') = g(b') = g(b) = c = q$.

Next we show $\text{dom}(\cdot) = \mathbb{N} \times \mathbb{N}$. Let arbitrary $p \in \text{dom}(\cdot)$. Then by definition there exists q such that $\langle p, q \rangle \in \cdot$. So there exists $a, b \in \mathbb{N}$ and c such that $\langle p, q \rangle = \langle \langle a, b \rangle, c \rangle$. So $p = \langle a, b \rangle \in \mathbb{N} \times \mathbb{N}$. Now let $p \in \mathbb{N} \times \mathbb{N}$. Then by definition there exists $a, b \in \mathbb{N}$ such that $p = \langle a, b \rangle$. Since $a \in \mathbb{N}$, as proven there exists $h : \mathbb{N} \rightarrow \mathbb{N}$ such that $\forall z (z \in h \leftrightarrow \exists m \in \mathbb{N} \exists x \in \mathbb{N} (z = \langle m, x \rangle \wedge \langle \langle m, a \rangle, x \rangle \in +))$, and there exists $g : \mathbb{N} \rightarrow \mathbb{N}$ such that $g(1) = a$ and $g \circ s = h \circ g$. By definition we then have $\langle \langle a, b \rangle, g(b) \rangle \in \cdot$ and $p = \langle a, b \rangle \in \text{dom}(\cdot)$.

For the range bound, let $q \in \text{ran}(\cdot)$. Then there exists p such that $\langle p, q \rangle \in \cdot$. So there exists $a, b \in \mathbb{N}$, $g : \mathbb{N} \rightarrow \mathbb{N}$, and c such that $\langle b, c \rangle \in g$ and $\langle p, q \rangle = \langle \langle a, b \rangle, c \rangle$. So $q = c = g(b) \in \mathbb{N}$.

Finally we verify the two properties of multiplication. Let arbitrary $m, n \in \mathbb{N}$. We first show $\cdot(\langle n, 1 \rangle) = n$. We know that $\langle \langle n, 1 \rangle, \cdot(\langle n, 1 \rangle) \rangle \in \cdot$. By definition there then exists $a, b \in \mathbb{N}$, $g : \mathbb{N} \rightarrow \mathbb{N}$ such that $g(1) = a$, and c such that $\langle b, c \rangle \in g$ and $\langle \langle n, 1 \rangle, \cdot(\langle n, 1 \rangle) \rangle = \langle \langle a, b \rangle, c \rangle$. So $\cdot(\langle n, 1 \rangle) = c = g(b) = g(1) = a = n$.
(next page)

Now we show $\cdot(\langle n, s(m) \rangle) = +(\cdot(\langle n, m \rangle), n)$. $s(m) \in \mathbb{N}$ exists, so we have $\langle \langle n, s(m) \rangle, \cdot(\langle n, s(m) \rangle) \rangle \in \cdot$. Then there exists $a, b \in \mathbb{N}$, $h : \mathbb{N} \rightarrow \mathbb{N}$ such that $\forall z_0 (z_0 \in h \leftrightarrow \exists m_0 \in \mathbb{N} \exists x \in \mathbb{N} (z_0 = \langle m_0, x \rangle \wedge \langle \langle m_0, a \rangle, x \rangle \in +))$, $g : \mathbb{N} \rightarrow \mathbb{N}$ such that $g(1) = a$ and $g \circ s = h \circ g$, and c such that $\langle b, c \rangle \in g$ and $\langle \langle n, s(m) \rangle, \cdot(\langle n, s(m) \rangle) \rangle = \langle \langle a, b \rangle, c \rangle$. We then have $\cdot(\langle n, s(m) \rangle) = c = g(b) = g(s(m)) = h(g(m))$ and $a = n$. Similarly we know that $\langle \langle n, m \rangle, \cdot(\langle n, m \rangle) \rangle \in \cdot$. So there exists $a_0, b_0 \in \mathbb{N}$, $h_0 : \mathbb{N} \rightarrow \mathbb{N}$ such that $\forall z_0 (z_0 \in h_0 \leftrightarrow \exists m_0 \in \mathbb{N} \exists x \in \mathbb{N} (z_0 = \langle m_0, x \rangle \wedge \langle \langle m_0, a_0 \rangle, x \rangle \in +))$, $g_0 : \mathbb{N} \rightarrow \mathbb{N}$ such that $g_0(1) = a_0$ and $g_0 \circ s = h_0 \circ g_0$, and c_0 such that $\langle b_0, c_0 \rangle \in g_0$ and $\langle \langle n, m \rangle, \cdot(\langle n, m \rangle) \rangle = \langle \langle a_0, b_0 \rangle, c_0 \rangle$. This means that $a_0 = n = a$, and that by our earlier statements we can assert that $h = h_0$, and $g = g_0$. So $\cdot(\langle n, m \rangle) = c_0 = g_0(b_0) = g(m)$. We then have $\cdot(\langle n, s(m) \rangle) = h(g(m)) = h(\cdot(\langle n, m \rangle))$. From the definition of h , there then exists $m_0, x \in \mathbb{N}$ such that $\langle \cdot(\langle n, m \rangle), h(\cdot(\langle n, m \rangle)) \rangle = \langle m_0, x \rangle$ and $\langle \langle m_0, a \rangle, x \rangle \in +$. This and the fact that $a = n$ allows us to conclude $+(\cdot(\langle n, m \rangle), n) = h(\cdot(\langle n, m \rangle)) = \cdot(\langle n, s(m) \rangle)$. $\square$

1.1.5 Properties of addition and multiplication on the naturals

Theorem 1.7. *Let $a, b, c \in \mathbb{N}$.*

1. *If $a + c = b + c$, then $a = b$ (Cancellation Law for Addition)*
2. *$(a + b) + c = a + (b + c)$ (Associative Law for Addition)*
3. *$1 + a = s(a) = a + 1$*
4. *$a + b = b + a$ (Commutative Law for Addition)*
5. *$a + b \neq 1$*
6. *$a + b \neq a$*
7. *$a \cdot 1 = a = 1 \cdot a$ (Identity Law for Multiplication)*
8. *$(a + b)c = ac + bc$ (Distributive Law)*
9. *$ab = ba$ (Commutative Law for Multiplication)*
10. *$c(a + b) = ca + cb$ (Distributive Law)*
11. *$(ab)c = a(bc)$ (Associative Law for Multiplication)*
12. *If $ac = bc$ then $a = b$ (Cancellation Law for Multiplication)*
13. *$ab = 1$ iff $a = 1 = b$*

(next page)

Proof. (1) We want to show

$$\forall a \in \mathbb{N} \forall b \in \mathbb{N} \forall c \in \mathbb{N} (a + c = b + c \rightarrow a = b)$$

To do this we define the set G as the set where

$$\forall c (c \in G \leftrightarrow c \in \mathbb{N} \wedge \forall a \in \mathbb{N} \forall b \in \mathbb{N} (a + c = b + c \rightarrow a = b))$$

(where existence follows from the subset axiom using $\mathbb{N}$ as the bounding set. Uniqueness follows from extensionality.) Then we show that $G = \mathbb{N}$. Showing $G \subseteq \mathbb{N}$ is straightforward; we need to show $1 \in G$ and $\forall g \in G (s(g) \in G)$. For the former, we already know $1 \in \mathbb{N}$. Let arbitrary $a, b \in \mathbb{N}$. Suppose that $a + 1 = b + 1$. By definition we then have $s(a) = s(b)$, so by injectivity $a = b$ and $1 \in G$.

Now the latter. Let arbitrary $g \in G$. As defined we then have $g \in \mathbb{N}$ and $\forall a \in \mathbb{N} \forall b \in \mathbb{N} (a + g = b + g \rightarrow a = b)$. $s(g) \in \mathbb{N}$ exists. Let arbitrary $a, b \in \mathbb{N}$ and suppose that $a + s(g) = b + s(g)$. Then by the definition of addition we have $s(a + g) = s(b + g)$ and, by injectivity, $a + g = b + g$. Since $g \in G$ we then have $a = b$. So $G = \mathbb{N}$. To prove the initial statement, let arbitrary $c \in \mathbb{N}$. Then from $\mathbb{N} = G$ the desired conclusion immediately follows. $\square$

Proof. (2) We want to show

$$\forall a \in \mathbb{N} \forall b \in \mathbb{N} \forall c \in \mathbb{N} ((a + b) + c = a + (b + c))$$

To do this we define G as the set where

$$\forall c (c \in G \leftrightarrow c \in \mathbb{N} \wedge \forall a \in \mathbb{N} \forall b \in \mathbb{N} ((a + b) + c = a + (b + c)))$$

(where existence and uniqueness can be easily shown) and show that $G = \mathbb{N}$. Showing $G \subseteq \mathbb{N}$ is straightforward. We need to show $1 \in G$ and $\forall g \in G (s(g) \in G)$.

For the former, we already know $1 \in \mathbb{N}$. Let arbitrary $a, b \in \mathbb{N}$. By the definition of addition we have

$$(a + b) + 1 = s(a + b) = a + s(b) = a + (b + 1)$$

Now the latter. Let arbitrary $g \in G$. Then by definition $g \in \mathbb{N}$ and $\forall a \in \mathbb{N} \forall b \in \mathbb{N} ((a + b) + g = a + (b + g))$. We know $s(g) \in \mathbb{N}$ exists. Let arbitrary $a, b \in \mathbb{N}$. By definition we then have

$$(a + b) + s(g) = s((a + b) + g) = s(a + (b + g)) = a + s(b + g) = a + (b + s(g))$$

So $\mathbb{N} = G$. To prove the initial statement, let arbitrary $c \in \mathbb{N} = G$. The desired conclusion immediately follows. $\square$

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Proof. (3)

□